Lecture 22

The principal series of $\mathrm{SL}(2,\mathbb{R})$

Objectives. *This lecture builds the principal series of $G = \mathrm{SL}(2,\mathbb{R})$. The principal series is induced from characters of the upper triangular subgroup. We realise it on functions on a circle. We prove unitarity when the parameter is purely imaginary. Then we compute how the Lie algebra moves vectors between K-types. Finally we use this ladder to decide which of these representations are irreducible.*

By the end of this lecture you will be able to:

- State the inducing data (ε, ν) and the role of the modular function.
- Pass between induced functions, homogeneous functions on the plane and the circle model.
- Prove that $\Pi_{\varepsilon,\nu}$ is unitary for $\nu \in i\mathbb{R}$.
- Compute the ladder of K-types, and prove that $\Pi_{\varepsilon,\nu}$ has a proper closed invariant subspace exactly at the points where the ladder breaks.

22.1 Inducing data

Throughout, $G = \mathrm{SL}(2,\mathbb{R})$. We use the matrices

$$k_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad a_t = \begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix}, \qquad n_x = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}.$$

Let $K = \{k_\theta\}$, $A = \{a_t\}$, $N = \{n_x\}$ and $M = \{\pm I\}$. The group M is the centre of G.

Definition 22.1. The *minimal parabolic subgroup* of G is the group $P = MAN$ of upper triangular matrices in G.

The subgroup P is the stabiliser of the line $\mathbb{R}(1,0)^{\mathsf{T}}$. So G/P is the set of lines through the origin of $\mathbb{R}^2$. This set is a circle, and it is compact. We induce from P for this reason: the functions below live on a circle, and their integrals over it are finite.

Each $p \in P$ has a unique form $p = ma_tn_x$. Its diagonal determines m and t; its upper-right entry then determines x. The subgroup N is normal in P, so $ma_tn_x \mapsto ma_t$ is a homomorphism. Note that $K \cap P = M$.

The tangent algebra of P has basis $H = \mathrm{diag}(1,-1)$ and $E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Direct multiplication gives

$$\mathrm{Ad}(a_t)H = H, \qquad \mathrm{Ad}(a_t)E = e^{2t}E, \qquad \mathrm{Ad}(n_x)H = H - 2xE, \qquad \mathrm{Ad}(n_x)E = E,$$

and $\mathrm{Ad}(-I) = \mathrm{id}$. Since Ad and det are multiplicative, these formulas give $|\det \mathrm{Ad}(ma_tn_x)| = e^{2t}$. We use the modular convention of Lecture 16, namely

$$\int f(xg)\,dx = \Delta(g)^{-1} \int f(x)\,dx.$$

With this convention $\Delta(g) = |\det \mathrm{Ad}(g)|^{-1}$. Therefore

$$\Delta_P(ma_tn_x) = e^{-2t}. \tag{22.1}$$

Up to the centre, P is the $ax + b$ group of Lecture 16 and Lecture 17 with $a = e^{2t}$, by Exercise 22.2. So this is its modular function $\Delta = 1/a$.

The group G itself is unimodular because it is connected with semisimple Lie algebra, by Lecture 16.

Definition 22.2. Let $\varepsilon \in \{0, 1\}$ and $\nu \in \mathbb{C}$. The *inducing character* $\chi_{\varepsilon,\nu} \colon P \to \mathbb{C}^\times$ is

$$\chi_{\varepsilon,\nu}(ma_tn_x) = \sigma_\varepsilon(m)\, e^{\nu t}, \qquad \sigma_\varepsilon(\pm I) = (\pm 1)^\varepsilon.$$

It is a continuous character, since it factors through $P \to MA$. It is unitary exactly when $\nu \in i\mathbb{R}$.

Definition 22.3. The space $I_{\varepsilon,\nu}$ consists of the continuous $f \colon G \to \mathbb{C}$ with

$$f(gp) = \Delta_P(p)^{1/2} \chi_{\varepsilon,\nu}(p)^{-1} f(g) \qquad (g \in G,\ p \in P).$$

For $g \in G$, the operator $\Pi_{\varepsilon,\nu}(g)$ on $I_{\varepsilon,\nu}$ is given by $(\Pi_{\varepsilon,\nu}(g)f)(h) = f(g^{-1}h)$ for $h \in G$.

By the modular formula (22.1), the defining condition reads

$$f(g\, ma_tn_x) = \sigma_\varepsilon(m)\, e^{-(\nu+1)t} f(g). \tag{22.2}$$

Left and right translations commute, so $\Pi_{\varepsilon,\nu}(g)$ preserves $I_{\varepsilon,\nu}$. Also $\Pi(g_1g_2) = \Pi(g_1)\Pi(g_2)$. We drop subscripts when (ε, ν) is fixed.

Remark 22.4. The factor $\Delta_P^{1/2}$ gives another form of the normalisation in Lecture 17. In Lecture 17 the square root appears in the action.

Let q be the rho-function of Lecture 17. It is a positive continuous function on G with $q(gp) = \Delta_P(p)q(g)$ for $g \in G$ and $p \in P$. Let f satisfy the unitary covariance condition of Lecture 17, and set $F(g) = \sqrt{q(g)}f(g)$. By the transformation law of q, the square root moves into the covariance condition. The action becomes plain left translation. For $\nu \in i\mathbb{R}$ the resulting condition is

$$|F(gp)|^2 = \Delta_P(p)|F(g)|^2.$$

This condition allows integration over G/P. Below we give the circle integral and prove its invariance directly. Nothing else from Lecture 17 is used.

The shift $e^{-t} = \Delta_P(a_t)^{1/2}$ is often written $e^{-\rho(tH)}$. Here ρ is the linear functional on $\mathbb{R}H$ with $\rho(H) = 1$, half the eigenvalue 2 of $\mathrm{ad}\, H$ on E. It is the analogue of the ρ of Lecture 9.

22.2 The plane model and the circle model

Let $\mathbf{e}_1 = (1,0)^{\mathsf{T}}$ and $u_\theta = (\cos\theta, \sin\theta)^{\mathsf{T}} = k_\theta \mathbf{e}_1$. Multiplying g on the right by n_x leaves the first column of g unchanged and adds a multiple of it to the second column. So a function with $f(gn_x) = f(g)$ depends only on the first column $g\mathbf{e}_1$, which can be any nonzero vector.

Lemma 22.5. *The map $g \mapsto g\mathbf{e}_1$ induces a homeomorphism $G/N \to \mathbb{R}^2 \setminus \{0\}$.*

Proof. If $g\mathbf{e}_1 = \mathbf{e}_1$, the first column of g is $\mathbf{e}_1$, and $\det g = 1$ forces $g \in N$. So the fibres of $g \mapsto g\mathbf{e}_1$ are the cosets gN. The map is onto, with continuous section

$$s(v) = \begin{pmatrix} v_1 & -v_2/|v|^2 \\ v_2 & v_1/|v|^2 \end{pmatrix}, \qquad s(v)\mathbf{e}_1 = v, \qquad \det s(v) = 1.$$

Thus $v \mapsto s(v)N$ is a continuous inverse. ∎

Proposition 22.6. *The rule $F(g\mathbf{e}_1) = f(g)$ is a bijection from $I_{\varepsilon,\nu}$ onto the continuous functions F on $\mathbb{R}^2 \setminus \{0\}$ satisfying*

$$F(cv) = \operatorname{sgn}(c)^{\varepsilon}\,|c|^{-(\nu+1)} F(v) \qquad (c \in \mathbb{R}^\times). \tag{22.3}$$

Under it, $\Pi_{\varepsilon,\nu}(g)$ becomes $F \mapsto F(g^{-1}\cdot)$.

Proof. Taking $p = n_x$ in the covariance law (22.2) shows $f(gn_x) = f(g)$. By the plane identification (Lemma 22.5), F is well defined, and $F = f \circ s$ is continuous. For $c = \pm e^t$ we have $g(\pm a_t)\mathbf{e}_1 = c\,g\mathbf{e}_1$. So the covariance law with $p = \pm a_t$ gives the homogeneity rule (22.3).

Conversely, given F, put $f(g) = F(g\mathbf{e}_1)$. Since $g\,m a_t n_x \mathbf{e}_1 = \pm e^t g\mathbf{e}_1$ for $m = \pm I$, the homogeneity rule gives the covariance law. Finally, $(\Pi(g)f)(h) = f(g^{-1}h) = F(g^{-1}h\mathbf{e}_1)$ for $h \in G$. So $\Pi(g)$ becomes $F \mapsto F(g^{-1}\cdot)$. ∎

A homogeneous function F is determined by its values on the unit circle. Put $\varphi(\theta) = F(u_\theta) = f(k_\theta)$. Then

$$F(ru_\theta) = r^{-(\nu+1)}\varphi(\theta) \quad (r > 0), \qquad \varphi(\theta+\pi) = (-1)^{\varepsilon}\varphi(\theta).$$

The second rule is the homogeneity rule (22.3) with $c = -1$. Let C_ε be the space of continuous 2π-periodic functions with this parity. For every ν, restriction to K identifies $I_{\varepsilon,\nu}$ with C_ε. This identification is the *compact picture*. In the compact picture the space C_ε does not depend on ν.

Restriction to K loses no data because of the Iwasawa decomposition of Lecture 20. It gives $G = KAN$, so $G = KP$. Every $g \in G$ has the form kp, and the defining condition gives $f(kp) = \Delta_P(p)^{1/2}\chi_{\varepsilon,\nu}(p)^{-1} f(k)$. So f is determined by its values on K.

Definition 22.7. For $h \in G$, the map τ_h of $\mathbb{R}/2\pi\mathbb{Z}$ is defined by $hu_\theta = |hu_\theta|\,u_{\tau_h(\theta)}$.

Apply the plane model (Proposition 22.6) with $g^{-1}u_\theta = |g^{-1}u_\theta|\,u_{\tau_{g^{-1}}(\theta)}$. This gives the circle model:

$$(\Pi_{\varepsilon,\nu}(g)\varphi)(\theta) = |g^{-1}u_\theta|^{-(\nu+1)}\,\varphi\big(\tau_{g^{-1}}(\theta)\big). \tag{22.4}$$

The parameter ν enters only through the scalar factor.

Figure 22.1 draws τ_h for $h = a_{1/2}$. The matrix h turns the unit circle into an ellipse, and each point of the ellipse is projected back to the circle along its ray. Directions crowd towards the horizontal axis.

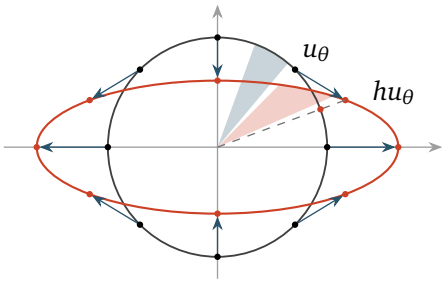


Figure 22.1: The map τ_h for $h = a_{1/2}$. Black dots are the points u_θ at multiples of $\pi/4$, and arrows lead to their images on the ellipse. The dashed ray projects $hu_{\pi/4}$ back to the circle at the angle $\tau_h(\pi/4)$. The thin sector of the disk and its image in the ellipse have equal area.

Example 22.8. For rotations $k_\alpha^{-1} u_\theta = u_{\theta-\alpha}$, so

$$(\Pi(k_\alpha)\varphi)(\theta) = \varphi(\theta - \alpha), \qquad \Pi(-I)\varphi = (-1)^\varepsilon \varphi.$$

The second formula uses $-I = k_\pi$. Thus the centre M acts by σ_ε for every ν. We call ε the central parity.

For the diagonal subgroup, $a_s^{-1} u_\theta = (e^{-s}\cos\theta,\ e^{s}\sin\theta)^{\mathsf{T}}$. Hence

$$r_s(\theta)^2 := |a_s^{-1} u_\theta|^2 = e^{-2s}\cos^2\theta + e^{2s}\sin^2\theta, \qquad \tan \tau_{a_s^{-1}}(\theta) = e^{2s}\tan\theta.$$

When $\varepsilon = 0$ the constant function $e_0 = 1$ lies in C_0, and $\Pi_{0,\nu}(a_s)e_0 = r_s^{-(\nu+1)}$.

22.3 Unitarity on the imaginary axis

Let $\mathcal{H}_\varepsilon$ be the closure of C_ε in $L^2(\mathbb{R}/2\pi\mathbb{Z})$, with inner product

$$\langle \varphi, \psi \rangle = \frac{1}{2\pi}\int_0^{2\pi} \overline{\varphi(\theta)}\,\psi(\theta)\,d\theta.$$

By Fourier analysis on the circle (Lecture 10), $\mathcal{H}_\varepsilon$ is the closed span of the functions $e_n(\theta) = e^{in\theta}$ with $n \equiv \varepsilon \pmod 2$. The key step towards unitarity is a change of variables on the circle.

Lemma 22.9. *Each τ_h is an orientation-preserving diffeomorphism. Moreover $\tau_{h_1 h_2} = \tau_{h_1} \circ \tau_{h_2}$ and*

$$\tau_h'(\theta) = |hu_\theta|^{-2}.$$

Proof. Linear maps commute with positive scaling, so $h_1 h_2 u_\theta$ is a positive multiple of $h_1 u_{\tau_{h_2}(\theta)}$. This gives the composition law, and $\tau_{h^{-1}}$ inverts τ_h. Locally $\tau_h(\theta)$ is a smooth branch of the argument of the nonvanishing smooth curve hu_θ. So τ_h is smooth.

For the derivative, write $r = |hu_\theta|$ and $\tau = \tau_h(\theta)$. Differentiate $hu_\theta = r\,u_\tau$ in θ:

$$hu'_\theta = r'u_\tau + r\tau' u'_\tau, \qquad u'_\theta = (-\sin\theta, \cos\theta)^{\mathsf{T}}.$$

Now compute the determinant of the pair (hu_θ, hu'_θ) in two ways. By multiplicativity of the determinant it equals $\det h \cdot \det(u_\theta, u'_\theta) = 1$. By the first display it also equals

$$\det(ru_\tau,\ r'u_\tau + r\tau' u'_\tau) = r^2\tau' \det(u_\tau, u'_\tau) = r^2\tau'.$$

Hence $\tau' = r^{-2} > 0$. The inverse $\tau_{h^{-1}}$ is smooth by the same argument. ∎

Figure 22.1 shows the reason. A thin sector of angle $d\theta$ in the unit disk has area $\frac{1}{2}d\theta$. Its image under h is, to first order, a sector of radius r and angle $d\tau$, of area $\frac{1}{2}r^2 d\tau$. Since $\det h = 1$, the matrix h preserves area, so $d\theta = r^2 d\tau$.

As a check, take $h = a_s^{-1}$ and differentiate $\tan\tau = e^{2s}\tan\theta$ from the diagonal example (Example 22.8):

$$\tau' = \frac{e^{2s}\sec^2\theta}{1 + e^{4s}\tan^2\theta} = \frac{e^{2s}}{\cos^2\theta + e^{4s}\sin^2\theta} = \frac{1}{e^{-2s}\cos^2\theta + e^{2s}\sin^2\theta} = \frac{1}{r_s(\theta)^2}.$$

Proposition 22.10. *For $\nu \in \mathbb{C}$, $\varphi, \psi \in C_\varepsilon$ and $g \in G$,*

$$\langle \Pi_{\varepsilon,-\bar\nu}(g)\varphi,\ \Pi_{\varepsilon,\nu}(g)\psi\rangle = \langle \varphi, \psi\rangle.$$

Proof. Put $h = g^{-1}$, $r(\theta) = |hu_\theta|$ and $\theta' = \tau_h(\theta)$. Since $r > 0$, we have $\overline{r^{-(-\bar\nu+1)}} = r^{\nu-1}$. By the circle-model formula (22.4), the integrand is

$$\overline{(\Pi_{\varepsilon,-\bar\nu}(g)\varphi)(\theta)}\,(\Pi_{\varepsilon,\nu}(g)\psi)(\theta) = r^{\nu-1}r^{-\nu-1}\,\overline{\varphi(\theta')}\psi(\theta') = \overline{\varphi(\theta')}\psi(\theta')\,r(\theta)^{-2}.$$

By the Jacobian lemma (Lemma 22.9), $r(\theta)^{-2}\,d\theta = d\theta'$. As θ runs once around the circle, so does θ', in the same direction. So the integral equals $\int_0^{2\pi} \overline{\varphi(\theta')}\psi(\theta')\,d\theta'$. After division by 2π, this is $\langle\varphi,\psi\rangle$. ∎

The Jacobian lemma rewrites the circle formula. With $\tau = \tau_{g^{-1}}$ we have $|g^{-1}u_\theta|^{-2} = \tau'(\theta)$, so

$$(\Pi_{\varepsilon,\nu}(g)\varphi)(\theta) = \tau'(\theta)^{(\nu+1)/2}\,\varphi(\tau(\theta)).$$

For $\nu = 0$ the factor is the square root of the Jacobian, and it keeps $|\varphi|^2 d\theta$ fixed when the circle is stretched. This is the square root of Lecture 17. For $\nu \in i\mathbb{R}$ the factor gains a phase of modulus one. For $\nu = -1$ the factor is 1, so φ transforms as a function. For $\nu = 1$ the factor is the Jacobian, so φ transforms as a density and its integral is invariant. Without $\Delta_P^{1/2}$, the two powers of r in the proof above would multiply to 1, and the Jacobian would be left over.

Theorem 22.11. *Let $\nu \in i\mathbb{R}$. Each $\Pi_{\varepsilon,\nu}(g)$ extends uniquely from C_ε to a unitary operator on $\mathcal{H}_\varepsilon$. The map $g \mapsto \Pi_{\varepsilon,\nu}(g)$ is a strongly continuous unitary representation of G.*

Proof. For $\nu \in i\mathbb{R}$ we have $-\bar{\nu} = \nu$. So the pairing identity (Proposition 22.10) says that $\Pi(g)$ preserves norms on C_ε. As C_ε is dense, $\Pi(g)$ extends uniquely to an isometry of $\mathcal{H}_\varepsilon$.

The relation $\Pi(g_1 g_2) = \Pi(g_1)\Pi(g_2)$ holds on C_ε by the plane model (Proposition 22.6). By continuity it holds on $\mathcal{H}_\varepsilon$. So $\Pi(g^{-1})$ inverts $\Pi(g)$. An invertible isometry is unitary.

For strong continuity, fix $\varphi \in C_\varepsilon$ and $g_0 \in G$. By the circle-model formula (22.4), the function $(g,\theta) \mapsto (\Pi(g)\varphi)(\theta)$ is continuous. It is uniformly continuous on $U \times [0, 2\pi]$ for a compact neighbourhood U of g_0. Hence $\sup_\theta |(\Pi(g)\varphi - \Pi(g_0)\varphi)(\theta)| \to 0$ as $g \to g_0$. The L^2 norm is at most this supremum.

For general $\psi \in \mathcal{H}_\varepsilon$ and $\delta > 0$, choose $\varphi \in C_\varepsilon$ with $\|\psi - \varphi\| < \delta$. Since every $\Pi(g)$ is unitary,

$$\|\Pi(g)\psi - \Pi(g_0)\psi\| \le 2\delta + \|\Pi(g)\varphi - \Pi(g_0)\varphi\|,$$

and the right side is below 3δ for g near g_0. ∎

Example 22.12. Unitarity can fail off the imaginary axis. For example, take $\varepsilon = 0$ and $\nu \in \mathbb{R}$. By the diagonal example (Example 22.8),

$$\|\Pi_{0,\nu}(a_s)e_0\|^2 = \frac{1}{2\pi}\int_0^{2\pi} \frac{d\theta}{(A\cos^2\theta + B\sin^2\theta)^{\nu+1}}, \qquad A = e^{-2s},\ B = e^{2s}.$$

For $A, B > 0$, the substitution $u = \tan\theta$ on each quarter period gives

$$\int_0^{2\pi} \frac{d\theta}{A\cos^2\theta + B\sin^2\theta} = 4\int_0^{\infty} \frac{du}{A + Bu^2} = \frac{2\pi}{\sqrt{AB}}.$$

Applying $-\partial_A - \partial_B$ to this identity gives

$$\int_0^{2\pi} \frac{d\theta}{(A\cos^2\theta + B\sin^2\theta)^2} = \frac{\pi(A+B)}{(AB)^{3/2}}.$$

Here $AB = 1$. For $\nu = 0$ the norm is 1, as the unitarity theorem (Theorem 22.11) predicts. For $\nu = 1$ the squared norm is $\frac{1}{2}(e^{2s} + e^{-2s}) = \cosh 2s$, which exceeds 1 for $s \neq 0$. Exercise 22.6 gives the general skew-symmetry test.

This shows only that the L^2 norm is not preserved. For $\varepsilon = 0$ and $0 < |\nu| < 1$, Lecture 23 uses a different inner product, which is preserved.

For general $\nu \in \mathbb{C}$ the substitution in the proof of the pairing identity (Proposition 22.10) still gives

$$\|\Pi_{\varepsilon,\nu}(g)\varphi\|^2 \le \max_\theta |g^{-1}u_\theta|^{-2\operatorname{Re}\nu}\,\|\varphi\|^2.$$

So each $\Pi_{\varepsilon,\nu}(g)$ is bounded, locally uniformly in g. The proof of the unitarity theorem (Theorem 22.11) then gives a strongly continuous representation on $\mathcal{H}_\varepsilon$. In its density step, 2δ is replaced by $2C\delta$, where C bounds $\|\Pi_{\varepsilon,\nu}(g)\|$ on a compact neighbourhood of g_0. By the pairing identity (Proposition 22.10), the adjoint is $\Pi_{\varepsilon,\nu}(g)^* = \Pi_{\varepsilon,-\bar{\nu}}(g^{-1})$.

22.4 K-types and the Lie algebra action

By the rotation example (Example 22.8), $\Pi(k_\alpha)e_n = e^{-in\alpha}e_n$. So the character $k_\alpha \mapsto e^{-in\alpha}$ of K occurs exactly once when $n \equiv \varepsilon \pmod 2$, and not at all otherwise. This matches the centre: $k_\pi = -I$ acts on e_n by $(-1)^n = (-1)^\varepsilon$. Let

$$V_{\varepsilon,\nu} = \operatorname{span}\{e_n : n \equiv \varepsilon \pmod 2\} \qquad \text{(finite sums)}.$$

For $\nu \in i\mathbb{R}$ this is the space of K-finite vectors of Lecture 21.

For smooth $\varphi \in C_\varepsilon$ and $X \in \mathfrak{g} = \mathfrak{sl}(2,\mathbb{R})$ put

$$d\Pi(X)\varphi = \frac{d}{ds}\Big|_{s=0} \Pi_{\varepsilon,\nu}(\exp sX)\varphi.$$

By the circle-model formula (22.4), the function $(s,\theta) \mapsto (\Pi(\exp sX)\varphi)(\theta)$ is smooth. So the difference quotients converge uniformly in θ. Hence they converge in $\mathcal{H}_\varepsilon$.

More generally, the function $(g,\theta) \mapsto (\Pi(g)\varphi)(\theta)$ is jointly smooth. On compact sets in a group chart, the difference quotients for its derivatives converge uniformly in θ. Hence φ is a smooth vector in the Hilbert representation, in the sense used in Lecture 21.

For $\nu \in i\mathbb{R}$ this derivative agrees with the derivative on smooth vectors in Lecture 15. We extend $d\Pi$ complex linearly to $\mathfrak{g}_\mathbb{C}$.

Lemma 22.13. *For smooth $\varphi \in C_\varepsilon$ and $X \in \mathfrak{g}$, using the dot product of $\mathbb{R}^2$,*

$$(d\Pi(X)\varphi)(\theta) = (\nu+1)\,(u_\theta \cdot Xu_\theta)\,\varphi(\theta) - \det(u_\theta, Xu_\theta)\,\varphi'(\theta).$$

Proof. Let $v(s) = \exp(-sX)u_\theta = r(s)u_{\tau(s)}$. Then $r(0)=1$, $\tau(0)=\theta$ and $v'(0) = -Xu_\theta$. As in the proof of the Jacobian lemma (Lemma 22.9), $\det(v,v') = r^2\tau'$, so $\tau'(0) = -\det(u_\theta, Xu_\theta)$. Also $(r^2)' = 2\,v\cdot v'$, so $r'(0) = -u_\theta \cdot Xu_\theta$. Now differentiate $r(s)^{-(\nu+1)}\varphi(\tau(s))$ at $s=0$. The product rule with these values of $r'(0)$ and $\tau'(0)$ gives the stated formula. ∎

We use the basis

$$W = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad Q = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

with $\exp(\alpha W) = k_\alpha$. We have $Wu_\theta = u_\theta'$, $Hu_\theta = (\cos\theta, -\sin\theta)^{\mathsf T}$ and $Qu_\theta = (\sin\theta, \cos\theta)^{\mathsf T}$. The derivative formula (Lemma 22.13) gives the table:

X	$u_\theta \cdot Xu_\theta$	$\det(u_\theta, Xu_\theta)$	$d\Pi(X)\varphi$
W	0	1	$-\varphi'$
H	$\cos 2\theta$	$-\sin 2\theta$	$(\nu+1)\cos 2\theta\,\varphi + \sin 2\theta\,\varphi'$
Q	$\sin 2\theta$	$\cos 2\theta$	$(\nu+1)\sin 2\theta\,\varphi - \cos 2\theta\,\varphi'$

Set $X_\pm = H \pm iQ \in \mathfrak{g}_\mathbb{C}$. Since $\sin 2\theta \mp i\cos 2\theta = \mp i e^{\pm 2i\theta}$,

$$d\Pi(X_\pm)\varphi = e^{\pm 2i\theta}\big((\nu+1)\varphi \mp i\varphi'\big).$$

The element W used here is the negative of the W in Lecture 21, so k_α here is $k_{-\alpha}$ there. In the notation of Lecture 21, $E = X_+/2$, $F = X_-/2$ and $H' = iW$. Thus the Fourier vector e_n has type n in that lecture's parametrisation, even though its character here is $e^{-in\alpha}$. These E and F are not the real matrices of § 22.1, where E is upper triangular.

Proposition 22.14. *For $n \equiv \varepsilon \pmod 2$,*

$$d\Pi(W)e_n = -in\,e_n, \qquad d\Pi(X_+)e_n = (\nu+1+n)\,e_{n+2}, \qquad d\Pi(X_-)e_n = (\nu+1-n)\,e_{n-2}.$$

In the letters of Lecture 21, $d\Pi(E)e_n = \tfrac{1}{2}(\nu+1+n)e_{n+2}$ and $d\Pi(F)e_n = \tfrac{1}{2}(\nu+1-n)e_{n-2}$.

Proof. Insert $e'_n = in\,e_n$ into the table row for W and into the display for $d\Pi(X_\pm)$. For example, $(\nu+1) - i(in) = \nu+1+n$. ∎

The ladder formulas also show that $d\Pi$ respects brackets, for every $\nu \in \mathbb{C}$. On e_n the eigenvalues of $d\Pi(W)$ give

$$[d\Pi(W), d\Pi(X_\pm)]e_n = \big(-i(n\pm 2) + in\big)(\nu+1\pm n)\,e_{n\pm 2} = \mp 2i\,d\Pi(X_\pm)e_n.$$

The products of the two ladder operators are

$$d\Pi(X_\mp)d\Pi(X_\pm)e_n = (\nu+1\pm n)(\nu-1\mp n)\,e_n = \big(\nu^2 - (1\pm n)^2\big)e_n.$$

Subtracting the two cases gives

$$[d\Pi(X_+), d\Pi(X_-)]e_n = \big((1+n)^2 - (1-n)^2\big)e_n = 4n\,e_n = 4i\,d\Pi(W)e_n.$$

In $\mathfrak{g}_{\mathbb{C}}$ the brackets $[W,H] = 2Q$, $[W,Q] = -2H$ and $[H,Q] = -2W$ give $[W, X_\pm] = \mp 2iX_\pm$ and $[X_+, X_-] = 4iW$ (Exercise 22.4). The elements W, X_+ and X_- span $\mathfrak{g}_{\mathbb{C}}$, and $d\Pi$ is complex linear. Hence $d\Pi([Y,Z]) = [d\Pi(Y), d\Pi(Z)]$ for all $Y, Z \in \mathfrak{g}_{\mathbb{C}}$. So $d\Pi$ is a representation of $\mathfrak{g}_{\mathbb{C}}$ on $V_{\varepsilon,\nu}$ for every $\nu \in \mathbb{C}$.

Proposition 22.15. *A subspace $U \subseteq V_{\varepsilon,\nu}$ is stable under $\Pi(K)$ and $d\Pi(\mathfrak{g})$ exactly when $U = \operatorname{span}\{e_n : n \in S\}$ for a set S such that the following two conditions hold. If $n \in S$ and $\nu+1+n \neq 0$, then $n+2 \in S$. If $n \in S$ and $\nu+1-n \neq 0$, then $n-2 \in S$. Hence the $(\mathfrak{g}, K)$-module $V_{\varepsilon,\nu}$ is irreducible if and only if ν is not an integer with $\nu \equiv \varepsilon + 1 \pmod 2$.*

Proof. Let U be K-stable and let $u = \sum c_n e_n \in U$ be a finite sum. The e_n with $c_n \neq 0$ span a finite-dimensional K-stable space F. The continuous curve $\alpha \mapsto e^{in\alpha}\Pi(k_\alpha)u$ lies in the closed subspace $U \cap F$ of F. Hence

$$\frac{1}{2\pi}\int_0^{2\pi} e^{in\alpha}\,\Pi(k_\alpha)u\,d\alpha = c_n e_n$$

lies in U. Thus U is spanned by the e_n it contains. Since H and Q are linear combinations of $X_\pm$, and $d\Pi(W)$ preserves every span of the e_n, stability under $d\Pi(\mathfrak{g})$ means stability under $d\Pi(X_\pm)$. By the ladder formulas (Proposition 22.14), this is the stated property of S.

If no coefficient $\nu+1\pm n$ vanishes, every nonempty admissible S contains all $n \equiv \varepsilon$. If $\nu+1+n_0 = 0$, then $S = \{n \le n_0\}$ is admissible and proper. If $\nu+1-n_0 = 0$, use $S = \{n \ge n_0\}$. Finally, $\nu = -(n_0+1)$ or $\nu = n_0 - 1$ with $n_0 \equiv \varepsilon$ means exactly that $\nu \in \mathbb{Z}$ and $\nu \equiv \varepsilon + 1 \pmod 2$. ∎

Figure 22.2 shows the criterion at four parameters.

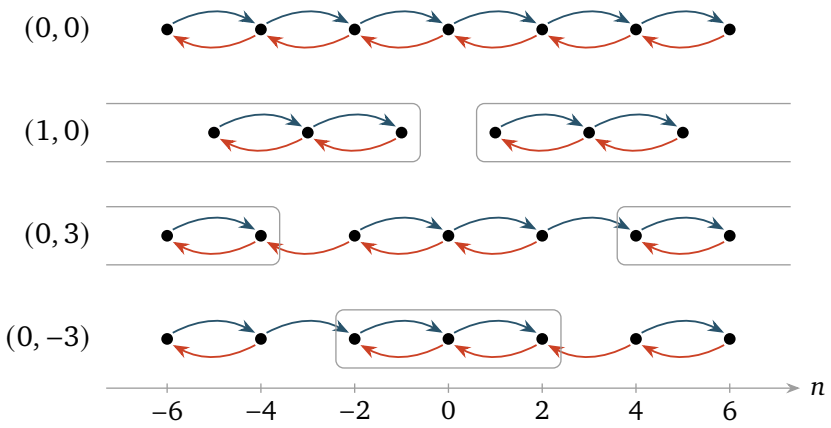


Figure 22.2: The ladder of $V_{\varepsilon,\nu}$, with rows labelled (ε, ν). Each dot is the one-dimensional space $\mathbb{C}e_n$; the rows continue in both directions. Upper arrows show $d\Pi(X_+)$ and lower arrows show $d\Pi(X_-)$. An arrow is missing where its coefficient vanishes. An invariant set of dots is one that no arrow leaves, and the boxes mark such sets.

22.5 Closed invariant subspaces

We now pass from the module $V_{\varepsilon,\nu}$ to the Hilbert representation on $\mathcal{H}_\varepsilon$. A $(\mathfrak{g}, K)$-submodule is an algebraic object and a closed invariant subspace is an analytic one, so the submodule criterion does not settle the question alone. In general the passage between them is Harish-Chandra's correspondence for admissible representations. For this family we prove what we need directly. First, a closed invariant subspace contains Fourier vectors. Next, a half-ladder has an invariant closed span.

Lemma 22.16. *Let $Y \neq 0$ be a closed subspace of $\mathcal{H}_\varepsilon$ invariant under $\Pi_{\varepsilon,\nu}(G)$. If $v \in Y$ and $\langle e_k, v\rangle \neq 0$, then $e_k \in Y$. If $e_k \in Y$, then $d\Pi(X_\pm)e_k \in Y$.*

Proof. By the rotation example (Example 22.8), the Fourier coefficient of $\Pi(k_\alpha)v$ at m is $e^{-im\alpha}\langle e_m, v\rangle$. So the integral

$$\frac{1}{2\pi}\int_0^{2\pi} e^{ik\alpha}\,\Pi(k_\alpha)v\,d\alpha = \langle e_k, v\rangle e_k$$

has Fourier coefficient $\langle e_k, v\rangle$ at k and 0 elsewhere. It is a norm limit of Riemann sums of vectors in Y, because the integrand is continuous. Since Y is closed, $e_k \in Y$.

The vector e_k is smooth, as shown before the derivative formula (Lemma 22.13). For $X \in \mathfrak{g}$, the quotient $t^{-1}(\Pi(\exp tX)e_k - e_k)$ lies in Y and converges in norm to $d\Pi(X)e_k$. So $d\Pi(X)e_k \in Y$. Complex combinations give $d\Pi(X_\pm)e_k \in Y$. ∎

Let $n \geq 1$ be an integer and let $\varepsilon \equiv n \pmod 2$. At $\nu = n - 1$ the ladder shows the half-ladder $k \geq n$. We define the closed span

$$H_n^+ = \overline{\operatorname{span}}\{e_{n+2m} : m \geq 0\} \subset \mathcal{H}_\varepsilon,$$

and likewise $H_n^- = \overline{\operatorname{span}}\{e_{-n-2m} : m \geq 0\}$. The ladder alone does not show that H_n^+ is invariant under the group, since $\Pi(g)$ is not a polynomial in $d\Pi$. We use power series in $z = e^{2i\theta}$. For a polynomial f put

$$(U_n f)(\theta) = e^{in\theta} f(e^{2i\theta}), \qquad \text{so that} \qquad U_n z^m = e_{n+2m}.$$

We use polynomials because a general vector of H_n^+ need not extend to a function holomorphic near the closed unit disk.

We also fix the complex matrix $\gamma = \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix}$. For $g \in G$ with $g^{-1} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, multiplying out gives

$$\gamma g^{-1} \gamma^{-1} = \begin{pmatrix} \alpha & \beta \\ \bar\beta & \bar\alpha \end{pmatrix}, \qquad \alpha = \tfrac{1}{2}\big(a + d + i(c - b)\big), \qquad \beta = \tfrac{1}{2}\big(a - d + i(b + c)\big). \tag{22.5}$$

The determinant gives $|\alpha|^2 - |\beta|^2 = ad - bc = 1$. So $|\beta| < |\alpha|$.

Proposition 22.17. *Let $n \geq 1$, $\varepsilon \equiv n \pmod 2$, $g \in G$, and let f be a polynomial. With $z = e^{2i\theta}$ and α, β as in (22.5),*

$$(\Pi_{\varepsilon,n-1}(g) U_n f)(\theta) = e^{in\theta}\,(\bar\beta z + \bar\alpha)^{-n}\, f\Big(\frac{\alpha z + \beta}{\bar\beta z + \bar\alpha}\Big). \tag{22.6}$$

Proof. For a real vector v we have $\gamma v = (v_1 + iv_2,\ v_1 - iv_2)^{\mathsf T}$. Hence $\gamma u_\theta = e^{-i\theta}(z, 1)^{\mathsf T}$. Write $g^{-1} u_\theta = r u_\tau$, with $r = |g^{-1} u_\theta|$ and $\tau = \tau_{g^{-1}}(\theta)$. Apply γ to both sides. Since $\gamma g^{-1} = (\gamma g^{-1} \gamma^{-1})\gamma$, this gives

$$r e^{-i\tau} \begin{pmatrix} e^{2i\tau} \\ 1 \end{pmatrix} = e^{-i\theta} \begin{pmatrix} \alpha z + \beta \\ \bar\beta z + \bar\alpha \end{pmatrix}.$$

The second entries give $\bar\beta z + \bar\alpha = r e^{i(\theta - \tau)}$. Dividing the first entries by the second gives $e^{2i\tau} = (\alpha z + \beta)/(\bar\beta z + \bar\alpha)$. Now the circle-model formula (22.4) with $\nu + 1 = n$ gives

$$(\Pi_{\varepsilon,n-1}(g) U_n f)(\theta) = r^{-n} e^{in\tau} f(e^{2i\tau}) = e^{in\theta}\big(r e^{i(\theta-\tau)}\big)^{-n} f(e^{2i\tau}),$$

which is the stated formula. ∎

Proposition 22.18. *Let $n \geq 1$ and $\varepsilon \equiv n \pmod 2$. The subspaces H_n^+ and H_n^- are invariant under $\Pi_{\varepsilon,n-1}$. Their orthogonal complements in $\mathcal{H}_\varepsilon$ are invariant under $\Pi_{\varepsilon,-(n-1)}$.*

Proof. The half H_n^+. Let $g \in G$, and let f be a polynomial. Since $|\beta/\alpha| < 1$, the geometric series

$$(\bar\beta w + \bar\alpha)^{-1} = \bar\alpha^{-1} \sum_{l \geq 0} (-\bar\beta w/\bar\alpha)^l$$

converges absolutely and uniformly on $|w| \leq 1$, and so do its powers. Therefore the function of w obtained from the right side of (22.6) has a power series $\sum_{l \geq 0} b_l w^l$ that converges uniformly on $|w| = 1$. Setting $w = e^{2i\theta}$, we see that $\Pi_{\varepsilon,n-1}(g) U_n f$ is a uniform limit, hence an L^2 limit, of finite combinations of the e_{n+2l}. So it lies in H_n^+. The vectors $U_n f$ span a dense subspace of H_n^+, and $\Pi_{\varepsilon,n-1}(g)$ is bounded. Hence $\Pi_{\varepsilon,n-1}(g) H_n^+ \subset H_n^+$ for every g.

The half H_n^-. For real ν the factor in (22.4) is real, so complex conjugation commutes with $\Pi_{\varepsilon,\nu}(g)$. Conjugation carries H_n^+ onto H_n^-. So H_n^- is invariant at $\nu = n - 1$.

The complements. By the adjoint formula after the pairing identity (Proposition 22.10), $\Pi_{\varepsilon,-(n-1)}(g)^* = \Pi_{\varepsilon,n-1}(g^{-1})$. Let $v \perp H_n^{\pm}$ and $u \in H_n^{\pm}$. Then

$$\langle u, \Pi_{\varepsilon,-(n-1)}(g)v\rangle = \langle \Pi_{\varepsilon,n-1}(g^{-1})u, v\rangle = 0.$$

■

Theorem 22.19 (Irreducibility; the two equivalences are cited). *Let $\nu \in \mathbb{C}$. The representation $\Pi_{\varepsilon,\nu}$ on $\mathcal{H}_\varepsilon$ has no closed invariant subspace other than 0 and $\mathcal{H}_\varepsilon$ exactly when $\nu \notin \varepsilon + 1 + 2\mathbb{Z}$. For $\nu \in i\mathbb{R}$, all $\Pi_{\varepsilon,\nu}$ are irreducible except $\Pi_{1,0}$. The space $\mathcal{H}_1$ of $\Pi_{1,0}$ is the orthogonal sum of the closed spans of $\{e_n : n > 0\}$ and $\{e_n : n < 0\}$, and each summand is irreducible.*
Moreover, for $\nu \notin \varepsilon + 1 + 2\mathbb{Z}$ we have $V_{\varepsilon,\nu} \cong V_{\varepsilon,-\nu}$ as $(\mathfrak{g}, K)$-modules. For $\nu \in i\mathbb{R}$, the representations $\Pi_{\varepsilon,\nu}$ and $\Pi_{\varepsilon,-\nu}$ are unitarily equivalent. These two equivalences are cited from Knapp, Representation Theory of Semisimple Groups*, §II.5, and Etingof's 18.757 notes, Lecture 19.*

Proof. We prove the first paragraph.

Irreducibility. Let $\nu \notin \varepsilon + 1 + 2\mathbb{Z}$, and let $W \neq 0$ be a closed invariant subspace. A nonzero vector of W has a nonzero Fourier coefficient, so $e_k \in W$ for some k, by Lemma 22.16. No coefficient $\nu + 1 \pm n$ with $n \equiv \varepsilon$ vanishes. So the ladder formulas (Proposition 22.14) and Lemma 22.16 put every e_n with $n \equiv \varepsilon$ in W. Hence $W = \mathcal{H}_\varepsilon$.

Reducibility. Let $\nu \in \varepsilon + 1 + 2\mathbb{Z}$. Then $\nu = n - 1$ or $\nu = -(n-1)$ for an integer $n \geq 1$ with $n \equiv \varepsilon \pmod 2$. By Proposition 22.18, H_n^+ or its orthogonal complement is a closed invariant subspace in the two cases. Each is nonzero and proper.

The imaginary axis. A number $\nu \in i\mathbb{R}$ lies in $\varepsilon + 1 + 2\mathbb{Z}$ only for $\nu = 0$ and $\varepsilon = 1$. So the exception is $\Pi_{1,0}$.

The point $(1,0)$. Take $n = 1$. Then H_1^+ and H_1^- are invariant, mutually orthogonal, and together they span $\mathcal{H}_1$. Let $W \neq 0$ be a closed invariant subspace of H_1^+. A nonzero vector of W has a nonzero coefficient at some odd $k \geq 1$, so $e_k \in W$. For odd $n \geq 1$ we have $1 + n \neq 0$, and for odd $n \geq 3$ we have $1 - n \neq 0$. So the ladder puts every e_n with odd $n \geq 1$ in W. Hence $W = H_1^+$. Complex conjugation gives the same result for H_1^-. ■

The pieces at the reducibility points return in Lecture 23.

The reducible module at $(\varepsilon, \nu) = (1, 0)$

The simplest reducibility point is $(\varepsilon, \nu) = (1, 0)$. The Fourier labels are the odd integers, and the ladder formulas become

$$d\Pi(X_+)e_n = (1+n)e_{n+2}, \qquad d\Pi(X_-)e_n = (1-n)e_{n-2}.$$

At $n = -1$, the upward coefficient vanishes. At $n = 1$, the downward coefficient vanishes. Thus the space $V_{1,0}$ is the direct sum

$$V_{1,0} = \operatorname{span}\{e_1, e_3, e_5, \ldots\} \;\oplus\; \operatorname{span}\{e_{-1}, e_{-3}, e_{-5}, \ldots\}.$$

Each summand is preserved by both ladder operators and by K. Within either chain, the remaining connecting coefficients are nonzero. The weight-projection argument in the

proof of the submodule criterion (Proposition 22.15) shows that each chain is algebraically irreducible.

The irreducibility theorem (Theorem 22.19) gives the two closed irreducible summands. Their group invariance is Proposition 22.18, which goes beyond the algebraic calculation above. Lecture 23 identifies them as limits of discrete series and explains their norms. The full odd ladder breaks at the gap between -1 and 1.

For comparison, $(\varepsilon,\nu)=(0,0)$ has only even labels. Neither equation $1+n=0$ nor $1-n=0$ has an even solution. Its algebraic module is therefore irreducible.

We compare the two modules through the Casimir element $\Omega = H^2+Q^2-W^2$ of $U(\mathfrak{g}_{\mathbb{C}})$. Exercise 22.4 shows $\Omega = X_+X_- - 2iW - W^2$ in $U(\mathfrak{g}_{\mathbb{C}})$. Since $d\Pi$ is a representation of $\mathfrak{g}_{\mathbb{C}}$, it extends to $U(\mathfrak{g}_{\mathbb{C}})$, and this identity holds for the operators. By the ladder formulas (Proposition 22.14), the three terms act on e_n by $\nu^2-(1-n)^2$, $-2n$ and n^2. Hence Ω acts on $V_{\varepsilon,\nu}$ by the scalar

$$\nu^2-(1-n)^2-2n+n^2=\nu^2-1.$$

By the dictionary of Lecture 21 above, $2EF+2FE=H^2+Q^2$ and $H'^2=-W^2$. So this Ω is the Casimir element $H'^2+2EF+2FE$ of Lecture 21.

Lecture 21 found $\Omega = m(m+2)$ on V_m. This equals ν^2-1 at $\nu=\pm(m+1)$. Indeed, take $\nu=-(m+1)$ with ε of the parity of m. For a function F on the plane, the homogeneity rule (22.3) then reads $F(cv)=c^mF(v)$. The homogeneous polynomials of degree m on $\mathbb{R}^2$ form a finite-dimensional invariant space. On the circle they span the e_n with $|n|\le m$ and $n\equiv m \pmod 2$.

At $(\varepsilon,\nu)=(1,0)$ and at $(0,0)$ this scalar is -1. The two modules nevertheless have different central parity and different reducibility. A Casimir value alone is not a classification label.

22.6 Self-checks

1. What is $\Pi_{\varepsilon,\nu}(-I)$?
   Answer. $(-I)^{-1}u_\theta = u_{\theta+\pi}$ has length 1. So $(\Pi(-I)\varphi)(\theta)=\varphi(\theta+\pi)=(-1)^\varepsilon\varphi(\theta)$. The centre acts by σ_ε for every ν.
2. For which (ε,ν) is e_0 fixed by G?
   Answer. We need $e_0\in C_\varepsilon$, so $\varepsilon=0$. We also need $d\Pi(X_+)e_0=(\nu+1)e_2=0$, so $\nu=-1$. Conversely, for $\nu=-1$ the factor in the circle-model formula (22.4) is 1, so $\Pi(g)e_0=e_0$.
3. Why is $V_{1,0}$ reducible?
   Answer. Here $\nu+1+n=0$ at $n=-1$ and $\nu+1-n=0$ at $n=1$. So $d\Pi(X_+)e_{-1}=0$ and $d\Pi(X_-)e_1=0$. The sets $\{n\ge 1\}$ and $\{n\le -1\}$ are admissible, and $V_{1,0}$ is the direct sum of their spans.

22.7 Summary

The principal series $\Pi_{\varepsilon,\nu}$ is induced from the character $\sigma_\varepsilon\otimes e^{\nu t}$ of $P=MAN$, shifted by $\Delta_P^{1/2}$. Restriction to K identifies every $I_{\varepsilon,\nu}$ with the continuous parity-ε functions on the circle. The group acts on these functions by the circle-model formula (22.4). The identity $\tau'_h=|hu_\theta|^{-2}$

makes the pairing of ν with $-\bar{\nu}$ invariant. Hence, for $\nu \in i\mathbb{R}$, the operators $\Pi_{\varepsilon,\nu}(g)$ extend to a strongly continuous unitary representation on $\mathcal{H}_\varepsilon$.

Each K-type occurring in $V_{\varepsilon,\nu}$ occurs once and is spanned by one e_n with $n \equiv \varepsilon \pmod 2$. The elements $X_\pm$ move n by ± 2 with coefficients $\nu + 1 \pm n$. We proved that the $(\mathfrak{g}, K)$-module $V_{\varepsilon,\nu}$ is reducible exactly at integers ν of parity opposite to ε. We also proved the same criterion for closed invariant subspaces of $\mathcal{H}_\varepsilon$. Fourier projection and the ladder give irreducibility away from these points. At $\nu = n - 1$ the power series of (22.6) show that the closed span of the half-ladder $k \geq n$ is invariant, and at $\nu = -(n-1)$ its orthogonal complement is. At $(\varepsilon, \nu) = (1, 0)$ the two half-ladders split $\mathcal{H}_1$ into irreducible summands. The module equivalence of $\pm\nu$ and its unitary version stay cited.

22.8 Further reading

- A. W. Knapp, *Representation Theory of Semisimple Groups: An Overview Based on Examples* (Princeton, 2001), §II.5 for $\mathrm{SL}(2,\mathbb{R})$, and §§VII.1–2 for induction from parabolic subgroups in general.
- P. Etingof, lecture notes for MIT 18.757, *Representations of Lie Groups*, Lectures 9 and 19.